



FOUR-COURSE SYSTEM

## COURSE 3 - DATA ENGINEERING 2026

# C3-054B CampusRides Lakehouse Gate

SEALED fresh full retest - explicit assignment only.

|              |                                                                                                        |
|--------------|--------------------------------------------------------------------------------------------------------|
| Artifact     | C3-054B                                                                                                |
| Status       | 2026 - LEARNER EDITION                                                                                 |
| Phase        | C3-04 Storage, File Formats & Lakehouse   13 lessons   34-46 focused hours                             |
| Prerequisite | C3-03 Gate PASS; Bridge warehousing/lakehouse mental models assumed, implementation depth taught here. |
| Brand        | Charcoal #1F2937   Teal #14B8A6   Soft Blue #3B82F6   AMOLED #000                                      |

House teaching rule: mental picture -> tiny example -> guided real file/table workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A file extension or successful write is never sufficient proof of correct table state.

# CampusRides scenario

CampusRides receives trip snapshots plus a correction batch. The correction updates one trip\_id, inserts one new row and adds fare\_class. This is a fresh full retest in a new domain.

**Gate B/C does not unlock merely because a prior Gate failed or remediation finished.**

## Independent rules

- Use supplied case ZIP; never edit fixtures to manufacture the expected control.
- Lesson/review/video answers are closed during the attempt; only permitted syntax references may be used.
- Declare business key/current-state grain and physical/logical table contract before implementation.
- Save real Parquet/Delta version/schema/MERGE/history evidence; do not substitute planned output for executed output.
- No destructive vacuum is required for the Gate.

## Required deliverables

- Choose and defend analytical file format plus partition strategy from workload/cardinality.
- Build current-state transactional table from base data.
- Apply correction with source uniqueness + grain-safe MERGE/upsert.
- Handle additive schema field deliberately and state incompatible-change policy.
- Prove key uniqueness plus source-to-target row/metric reconciliation.
- Read a prior committed version and explain exact change.
- Measure file health; define compaction trigger + post-rewrite parity.
- Define retention window preserving required recovery/time travel.
- Compare Delta and Iceberg conditionally across at least three decision factors.
- Provide table contract + maintenance/recovery runbook.

## Competency Gate - every required competency must pass

| Competency                                                           | PASS evidence                                                         |
|----------------------------------------------------------------------|-----------------------------------------------------------------------|
| Physical layout / format / partition reasoning                       | Observed, validated, reproducible and explainable on this fresh case. |
| Real Parquet metadata and file-health evidence                       | Observed, validated, reproducible and explainable on this fresh case. |
| Transactional Delta version/log evidence                             | Observed, validated, reproducible and explainable on this fresh case. |
| Controlled schema evolution and contract change                      | Observed, validated, reproducible and explainable on this fresh case. |
| Grain-safe MERGE/upsert with reconciliation                          | Observed, validated, reproducible and explainable on this fresh case. |
| Historical version read and retention reasoning                      | Observed, validated, reproducible and explainable on this fresh case. |
| Layer/catalog/open-table-format architecture and maintenance handoff | Observed, validated, reproducible and explainable on this fresh case. |

**PASS only if every required competency passes AND zero critical failures remain. Every required competency stands on its own; one failure blocks PASS.**



## Evidence packet

| Evidence                         | Your saved proof/path/note |
|----------------------------------|----------------------------|
| Format/layout decision           |                            |
| Parquet metadata/health evidence |                            |
| Delta version/log evidence       |                            |
| Schema-change record             |                            |
| MERGE/upsert metrics + key check |                            |
| Row/metric reconciliation        |                            |
| Historical version comparison    |                            |
| Layer/catalog/table contract     |                            |
| Compaction + retention plan      |                            |
| Open-table-format decision       |                            |
| Recovery/maintenance runbook     |                            |